

Can We Talk?: The Impact of Willingness to Recommend on a New-to-Market Service Brand Extension Within a Social Network

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Abstract

Adoption of new services in the marketplace, and the impact that word of mouth (WOM) has on adoption, has long been of interest to marketers. Managers have therefore become increasingly interested in measuring WOM activity most commonly through the recommend intention metric. The circumstances under which the predictive ability of this metric can be established, however, are not clear. This research provides the first longitudinal examination of the relationship between recommend intention and the adoption of a new-to-market service (NTMS) brand extension. Analysis is conducted using anonymized data provided by a large U.S. telecommunications provider for 791 customers and their corresponding telephone network (11,552 individuals). The findings indicate an interaction effect where recommend intention predicts new service adoption only when the recommending customers are more recent adopters of the service and are in more frequent contact with the potential customer. Therefore, when managers are using the recommend intention metric to predict adoption, there is a need to take into consideration the exposure of the individual to others in their network and the timing of their adoption.

Keywords

social network, word of mouth, service adoption, willingness to recommend, customer satisfaction

Introduction

The adoption of new services by consumers is perhaps one of the most important research problems in marketing. It is particularly important as services constitute a significant majority of most economies and the volume of new services being introduced into the market is large. Many of the new products being introduced into the market, however, are extensions of existing brands. Research has found that 80% of new product introductions take the form of extensions rather than new-to-market products (Taylor 2004). In the services domain, exemplars of such extensions include the introduction of new financial instruments by insurance companies, a computer repair provider offering data backup and storage services, signing up for a financial services firm's online banking, or "alert" service and offering Internet connectivity services by cable companies. Such extensions can be targeted at both current customers who own at least one but not all services the company offers or potential customers who have not yet purchased from the company.

Research indicates that interpersonal communication (usually referred to as "social contagion" or "word of mouth") is one of the most powerful means to increase adoption of new products and services by both current customers and potential customers (Barlyn 2007; Chevalier and Mayzlin 2006; Danaher and Rust 1996; Godes and Mayzlin 2004, 2008; Hogan, Lemon, and Libai

2003; Kumar, Peterson, and Leone 2007; Rust and Chung 2006; Silverman 2001). Since consumer-to-consumer word-of-mouth (WOM) communication is an important aspect of marketing and a major driver of success in the marketplace, it has become the impetus for many viral marketing and buzz strategies. As a result WOM marketing efforts and organizations have also seen corresponding interest and growth (Vranica 2006; WOMMA 2007). In fact, WOM marketing is reported to be the fastest growing sector in marketing and media services with revenues expected to break \$3.7 billion in 2011 (PQ Media 2007).

Therefore, researchers and managers alike have become increasingly interested in measuring and managing WOM activity. In particular, managers have typically sought to identify customers who are believed likely to spread positive WOM about their firms' products and services and measure their

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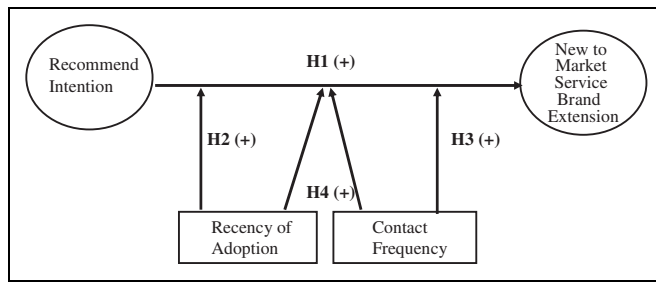


Figure 1. The conceptual model.

intentions to talk about their companies' products. The most popular and widespread metric used to measure and predict such WOM activity is recommend intention. This is in part because stated recommend intention is typically easier to collect than actual WOM behavior, particularly if the WOM is not within the realm of a company initiated, sponsored, and tracked referral or WOM program. More importantly, recommend intention has become widely adopted after the introduction and rise in popularity of the Net Promoter metric (Reichheld 2003, 2006). As such, recommend intention is presumed to be a good predictor of new product adoption by individuals within the social networks of a firm's customers.

Recent research, however, has called these claims into question and found recommend intention to either have no relationship or a weak relationship to company sales performance (Keiningham et al. 2007a; Morgan and Rego 2007; Sharp 2008). Therefore, it is not clear to what extent recommend intention is a valid predictor of adoption. To date, there is no scientific research that examines the ability of and circumstances under which recommend intention can aid in understanding adoption by individuals within a social network. The question then is *when* and *to what extent* is recommend intention a good predictor of actual new product adoption? Given that (a) the role WOM plays in new service adoption is crucial, (b) WOM marketing is a growing trend, and (c) the recommend intention metric is the most widely used metric to predict actual WOM, it is important to determine the conditions under which recommend intention links to actual new service adoption. Furthermore, despite the importance and centrality of WOM to marketing practice, and the large amount of WOM (or social contagion) research in marketing and in other fields (e.g., sociology, physics, and economics), surprisingly little is known about the drivers of successful individual WOM transmission (Stephen and Lehmann 2009).

This research uses a telecommunications services context to explain the propensity of existing customers to get potential customers in their network to adopt a new service brand extension by introducing two individual-level moderator variables. These two proposed variables are (a) recency of adoption of the new service by the person transmitting WOM to others in their network, who have not adopted the new service yet and (b) frequency of interactions between the person transmitting the WOM and individuals in their network (see Figure 1). This research represents the first examination of the relationship

between the recommend intention metric and actual new service brand extension adoption and two individual-level moderators using a longitudinal dataset within a live discussion social network.

The article is organized in the following way: First, we present our theoretical background and hypotheses about these relationships. Next, we provide details of our data and survey approach and elaborate on the methodological approach. Our investigation uses anonymized data from a large U.S. telecommunications provider to investigate the proposed relationships. The data consist of 791 customers of a core brand service and a new-to-market service (NTMS) brand extension, and each of these customers' corresponding contacts within his or her telephone network who use the firm's core service but not the extension (a total of 11,552 individuals). Finally, after we present and discuss our empirical results, we end with conclusions, managerial implications, limitations, and issues for further research.

Theoretical Background

Following the pioneering work of Rogers (1962) and Bass (1969), much of new product diffusion is believed to result from social contagion (i.e., "actors' adoption behaviors are a function of the exposure to other actors' knowledge, attitudes, or behaviors concerning the innovation," Van den Bulte and Wuyts 2007, p. 39). Van den Bulte and Stremersch (2004, p. 530) note that "researchers have offered different theoretical accounts of social contagion, including learning under uncertainty, social-normative pressures, competitive concerns, and performance network effects. Although these contagion mechanisms are conceptually distinct, their expressions in diffusion data of a single innovation are often indistinguishable." Although there are opposing findings (Burt 1987; Strang and Tuma 1993; Van den Bulte and Lilien 2001), it is generally believed that WOM communications shape consumers' attitudes and behavior toward product offerings (Arndt 1967; Brown and Reingen 1987; Danaher and Rust 1996; Herr, Kardes, and Kim 1991; Wangenheim and Bayón 2004), ultimately affecting customer equity. Rust, Zeithaml, and Lemon (2000, p. 46) observe, "the effect [of word-of-mouth] is notoriously hard to measure, but it is frequently significantly large." In fact a recent study (Villanueva, Yoo, and Hanssens 2008) finds that marketing-induced customers add more short-term value, but WOM customers add nearly twice as much long-term value to the firm. WOM has therefore become an important component in sociologists' models of social contagion and is found to facilitate the flow of information in social networks (Frenzen and Nakamoto 1993).

Link Between Recommend Intention Measure and New Service Adoption

As a result of the interest in the *effects* of WOM communications, managers, and researchers have begun focusing attention on metrics that provide insight into customers' WOM behavior.

Zeithaml et al. (2006) note that managers want “headlight” or forward-looking customer metrics that ... allow firms to anticipate changes and provide opportunities to increase the value of the customer base.” To do this, managers frequently seek to anticipate customers’ future behaviors based upon behavioral intentions (Gupta and Zeithaml 2006). The most commonly surveyed behavioral intention metrics for transmission of information between individuals is WOM intention (e.g., recommend intention). The underlying belief is that this metric will be positively correlated with customers’ actual WOM behavior and in turn link to service adoption behavior.

This positive relationship is expected to exist for several reasons. First, WOM is commonplace and individuals are motivated to engage in WOM for a variety of reasons such as the desire to help others, share experiences with others, and promotion of self-concept and product involvement (Feick and Price 1987; Mangold, Miller, and Brockway 1999) among others. WOM generally speaking comes from people we know and tends to be considered more reliable and trustworthy than recommendations we get through more formal marketing channels. Also, unlike advertising, WOM is also backed up by social pressures to conform making it even more effective. Hence WOM is considered an influential source and mechanism in service adoption.

Prior theory has also proposed links between satisfaction, WOM, and new customer acquisition. Such a link is central to models such as the Service Profit Chain (Heskett et al. 1994), Return on Quality (Rust et al. 1995), and Satisfaction Profit Chain (Anderson and Mittal 2000) where WOM is recognized as both a consequence of customer satisfaction and an antecedent to revenue and profit due to new customer acquisition. Satisfaction has been linked to customers’ likelihood to recommend the product and in turn to adoption in many contexts (e.g., Wangenheim and Bayon 2007).

Finally, it is important to establish to what extent recommend intention moves in tandem with actual behavior, that is, whether customers who say they will recommend actually do. There is evidence from research confirming the link between WOM intention and actual WOM behavior. Comparing different customer behavior metrics, Keiningham et al. (2007a, 2007b) find that WOM intention has the strongest correlation ($r = 0.40$) to actual recommend behavior.

As a result of this link between WOM and adoption, recommend intention represents one of the most watched and tracked metrics among managers. This is in large part the result of the popularity of the Net Promoter movement, which touts recommend intention as the “ultimate question” for linking to firm performance (Reichheld 2003, 2006). To calculate this Net Promoter score, survey respondents are asked to rate their likelihood of recommending a company on a 0–10 point scale. The proportion of respondents rating the firm a 6 or less (called “detractors”) is subtracted from the proportion of respondents rating the firm a 9 or 10 (called “promoters”); this difference represents a firm’s Net Promoter score.

Recent research, however, has contradicted these claims and frequently found no relationship or very weak relationships

between recommend intention and sales revenue (Keiningham et al. 2007; Morgan and Rego 2007; Pingitore et al. 2007). Kumar, Peterson, and Leone (2007) report similar findings, being unable to link recommend intention and actual sales in the telecom and financial services industries.

It is not clear therefore, whether WOM intentions translate to actual recommendations and ultimately new product adoption. Nonetheless, as described above, there is a large body of research that finds a positive link between recommend intention and adoption.

We further argue that a positive link between recommend intention and adoption is expected to be even stronger in the case of brand extensions compared to new services. Brand extensions are extensively used as a means of introducing new services into the market because positive associations from the existing brand are more easily transferred on to the brand extension carrying the same brand, provided the parent brand is of a certain quality and there is perceived fit between an extension and the parent brand (Aaker and Keller 1990). There are also potential cost and distribution efficiencies associated with this strategy, and the risk of failure is controlled to a greater extent compared to a completely NTMS introduction. Therefore, we would expect WOM by an individual who already owns a particular brand to have a stronger link to the adoption of a brand extension when compared to a completely new service. In the case of our investigation, this research focuses on the adoption of a brand extension by customers of a telecommunications provider who use the firm’s core service but not the extension; therefore we would expect respondents in our sample to be more positively predisposed to the effects of WOM on the adoption of the brand extension.

Since the research suggests the ability of recommend intention to aid in the prediction of actual adoption, we propose the following hypothesis (see Figure 1):

Hypothesis 1: Recommend intention will be positively associated with adoption of a new service by existing customers.

Moderating Effect of Adoption Recency on the Recommend Intention—New Service Adoption Link

The idea that the recency of a purchase would play a key role in customers’ future behaviors is both intuitive and supportable using current marketing theory. It is based on the assumption that (a) recent events are more novel, salient in memory, are more accessible, easily recalled and hence have more influence on current decisions and (b) recency is simply a measure of customer engagement. Timing and sequence of events for example has been found to be an important predictor of how a customer experiences a service. Verhoef, Antonides, and DeHoog (2004) confirm the peak-end rule (Kahneman, Wakker, and Sarin 1997) by demonstrating the importance of the most recent experience in evaluation of service encounters. They find that average performance during the encounter is important but that peak end performance (i.e., extreme

positive) tends to be critical to customers evaluations of service performance.

Research also indicates that recency of purchase of a particular product/service affects WOM activity. The finding regarding the effect of recency of purchase however poses disparate and diverging results. Recent research by Iyengar, Van den Bulte, and Valente (2010) finds that customers that are recent have much less influence compared to longer term customers. On the other hand, Godes and Mayzlin (2009) find that less loyal customers (more recent adopters of a product) are likely to be more influential in affecting sales. Other studies corroborate Godes and Mayzlin's (2009) findings. In a large-scale study of fixed line telephone users in the United Kingdom, Ranaweera and Prabhu (2003) found that new customers are more likely to engage in both positive and negative WOM than are long-term customers. Recent customers are also more likely to transmit WOM information due to the enthusiasm that comes with initial choice, which can be thought of as excitement about, and positive attitude toward, a given product. And similar to the viral nature of new information, enthusiasm can be contagious as well (Stephen and Berger 2009). People who are more enthusiastic about a product should be more likely to talk about it, speak positively when they talk, and consume it, which in turn should influence adoption by the people receiving the communication. Hill, Provost, and Volinsky (2006) found that consumers who had communicated with an early adopter of a telecommunications service were three to four times more likely to respond to an offer for the product. Furthermore, as the new service context under investigation is a NTMS brand extension, the customer engaging in WOM is likely to be more knowledgeable and hence potentially more persuasive when communicating.

Naturally, as more time passes, enthusiasm is more likely to abate. This is driven in part by our inherent desire for novelty. Researchers in psychology and marketing find that as time goes by, the magnitude of hedonic response caused by exposure to a stimulus tends to decrease, a process referred to as habituation or adaptation (Fournier and Mick 1999; Thompson and Spencer 1966). Sensory adaptation (i.e., we learn to ignore stimuli to which we have prolonged exposure) constitutes a form of habituation with which we are all familiar. For example, we tend to no longer consciously hear background noise after brief exposure. The timing the stimulus is encountered therefore becomes important. The more recent the event, the less likely habituation and adaptation are to occur. Recent experiences are more noticeable, vivid in memory and salient and as a result more likely to be talked about.

Based on the findings of the literature regarding the widespread use of recency as a measure, the effect of recency, augmented with the motivation that our inherent enthusiasm and excitement for something "new" provides and the time needed for habituation and adaptation to set in, we would expect more recent adopters of a new service to engage in more WOM and be more influential in getting potential customers in their network to adopt new services. We therefore hypothesize the following (see Figure 1):

Hypothesis 2: Recommend intention will demonstrate a stronger relationship to adoption of a new service by existing customers when current users are more recent adopters of the new service.

Moderating Effect of Contact Frequency on the Recommend Intention—New Service Adoption Link

On an individual level, there could be a variety of reasons why one engages in conversations with others. Overall, it can be described as a social process that provides benefits and has consequences such as the creation of social relationships (i.e., initiation and creation of ties in their social networks). Similarly, transmitting WOM can also fulfill certain social needs (e.g., attracting attention from others, acquiring helpful information from others, strengthening friendships, and feeling that one's opinions have been "heard"). The closeness of the ties of people in the network, however, can differ to a great extent. As such social ties vary in terms of their strength (e.g., "strong" for friends and "weak" for acquaintances) and affect the opportunities for information to be shared or exchanged. For instance, Brown and Reingen (1987) and Reingen and Kernan (1986) found that while strong social ties are more likely to be used, weak ties are important for bridging gaps in a social network. In fact Godes and Mayzlin (2009) find that for a product with a low initial awareness level, WOM that is most effective at driving sales is created by less loyal (not highly loyal) customers and occurs between acquaintances (not friends). Cellular automata (or agent-based) simulations of WOM over strong and weak ties find similar results (Goldenberg, Libai, and Muller 2001), as does the "strength of weak ties" literature in sociology.

De Bruyn and Lilien (2008) found that characteristics of the social tie influence recipients' behaviors but have different effects at different stages: tie strength facilitates awareness and perceptual affinity triggered recipients' interest making them more likely to be stronger advocates. In general, it seems that people have a preference for talking to strong tied friends ("strong tie bias") when sharing information and are willing to share valuable information about limited (scarce) deals with strong ties (friends) but not weak ties (acquaintances; Frenzen and Nakamoto 1993).

Intuitively, length of contact (number of contacts and especially duration of contacts) in a social network is indicative of the closeness of relationships and such close relationships have the potential to influence choice. For example, Bandiera and Rasul (2006) found that farmers' decisions to adopt a new crop are influenced by the adoption choices of farmers in their "in-group" social network of family and friends. Conley and Udry (2005) also found similar effects for farmers in Ghana deciding to adopt and use a new farming technology. Bell and Song (2007) found similar significant contagion effects (based on geographic proxies) for the adoption of an Internet grocery store at the zip code level.

The frequency and duration of contact would also be expected to impact exposure to WOM recommendations for

a product or service where more frequent interactions would be accompanied with greater exposure to WOM communications creating the potential for increased impact. The effect of repeated exposure has been widely investigated by researchers especially in the advertising domain (e.g., Batra and Ray 1986; Belch 1982; Burke and Srull 1988; Calder and Sternthal 1980). Researchers for instance find that repetition of an advertising message can increase recall (Belch 1982; Burke and Srull 1988; Cacioppo and Petty 1979).

While research into the effects of repeated exposure to an advertising message has found an inverted U relationship between exposure and impact, these results are not applicable to WOM advertising for a number of reasons. Even though the positive benefits associated with early stages of multiple exposure to a message reach a saturation point at some point and thereafter repeated exposure results in negative attitudes (due to habituation—acclimation by frequent repetition or prolonged exposure and tedium—monotony and boredom resulting from prolonged exposure; Berlyne 1970; Cacioppo and Petty 1979; Stang 1975), this is less applicable to the case of WOM advertising. We would expect consumers to be much less likely to reach such a saturation point because one-on-one WOM is a social phenomenon and is typically transmitted as part of a broader interactive dialogue rather than a targeted one-way message as is the case in advertising. Furthermore, due to the two-way nature of conversations, as such social cues will often become apparent in the interaction should the recipient of the information reach a level of discomfort with WOM advertising.

While we recognize that the research of Godes and Mayzlin (2009) and Goldenberg, Libai, and Muller (2001) appears to argue in favor of weak ties over strong ties in the WOM–product adoption link, we believe that in line with researchers Bandiera and Rasul (2006) and Conley and Udry (2005), strong ties would be expected to have greater predictive significance in the repurchase intention–service adoption link. Our argument is based in large part on the exposure effect—in particular, in a telephone-based social network, strong links have greater communication duration, and therefore are likely to provide greater exposure to WOM advertising communication. Furthermore, since the new service context in this research is a NTMS brand extension of the core service (to which both the transmitter and the recipient of the WOM communication subscribe), we would expect the effect of exposure to WOM to be exacerbated due to increased relevancy (compared to an unrelated brand/new service). As a result, we hypothesize a general “strong tie bias” among transmitters of product-related information. In particular, we hypothesize (see Figure 1):

Hypothesis 3: Recommend intention will demonstrate a stronger relationship to adoption of a new service by existing customers when current users and potential users in a social network spend more time communicating with one another.

Interaction Effect of Contact Frequency and Adoption Recency on the Recommend Intention—New Service Adoption Link

Building upon Hypotheses 2 and 3, we would expect to find an even stronger effect of frequency of contact when the WOM transmitter has also made the purchase more recently. Therefore, when a customer is a more recent adopter *and* is in more frequent contact with others in his or her network, we would expect that the effect of recommend intention on service adoption to increase, that is, the link from intention to actual service adoption should be stronger. In fact, we argue that there should be an interaction between the two that is greater than the sum of the combination of their individual components.

Other researchers have found moderating effects to be important in the study of service adoption. In their investigation of how satisfaction and payment equity (defined as the perceived fairness of the price) affect cross-buying at a multiservice provider, Verhoef, Hoekstra, and Frances (2001) find that the effect of satisfaction differs between customers with lengthy and short relationships. But they also find that payment equity negatively affects cross-buying for customers with long relationships. However, if the prices of the supplier are perceived as fairer than the prices of the competitor, the customers’ probability of cross-buying increases, demonstrating moderating effects. Research by Wu and Yen (2007) demonstrated that interaction effects influence customers’ evaluation of brand extensions. In particular, their research examined the interaction effects of parent brand strength on various aspects of brand extension breadth and similarity on consumers’ acceptance of a brand.

The argument for the concomitant effect of adoption recency and contact frequency can easily be understood by thinking about the underlying reasoning behind why we would in part expect each component to have an impact on service adoption. Frequency and duration of contact would be expected to have a direct impact on *exposure* to WOM advertising. The newness (i.e., recency) of purchase by the WOM transmitter is expected in part to have an impact on the *enthusiasm* associated with the WOM advertisement (Stephen and Berger 2009). This enthusiasm will clearly affect the believability and persuasion of the WOM advertisement.

Simplistically, imagine contact frequency as being equal to exposure and recency of purchase as being equal to believability. In such a scenario, each component is not simply complementary but is instead essential in affecting service adoption through WOM advertising. If exposure is optimal but believability is zero, then the effect of WOM advertising on service adoption will be severely limited. And if believability is high but there is no exposure, then there can be no effect of WOM advertising. In other words, it is useless to have an audience without a persuasive WOM message and it is equally useless to have a great WOM message with no one to listen to it. The nature of these two components is therefore quite complementary.

Consequently, among more recent adopters of the NTMS brand extension, recommend intention should have a

Table 1. Survey Household Demographic Information

	Percent		Percent
Education		Age	
Some high school	1.8	Under 18	0.1
High school	16.4	18–24	2.4
Some college	24.3	25–34	15.1
College degree	29.4	35–44	23.6
Trade school	5.4	45–54	27.9
Some graduate work	4.7	55–64	17.2
Graduate degree	18	65 or older	13.8
Household income		Ethnicity	
Under \$25000	12	White/Caucasian	78.1
Between \$25000 and \$40000	19.1	African American	11.2
Between \$40000 and \$55000	17.1	Hispanic or Latin American	2.3
Between \$55000 and \$70000	15	Native American	0.9
Between \$70000 and \$85000	12.1	Asian American or Pacific Islander	1.5
Between \$85000 and \$100000	8.3	Some other race	2.7
Over \$100000	16.6	Refused	3.4
Gender			
Male	39.1		
Female	60.9		

significant effect on adoption rates when there is a higher frequency of communication between current and potential customers. Therefore, we propose the following hypothesis (see Figure 1):

Hypothesis 4: Recommend intention will demonstrate a stronger relationship to adoption of a new service by existing customers when current users are more recent adopters of the new service AND current and potential users in a social network spend more time communicating with one another.

Method

Data Collection and Measurement

Our analysis focuses on the relationship between the WOM intentions of those households who already have the NTMS brand extension and the subsequent subscription rates to this service among those customers who are in each household's telephone network. The service extension is a high-speed connectivity service, with at least one comparable service available from a competitor in each market studied. At the time of the data collection stage, the service extension had been widely available for several years, and the core service was required to obtain the NTMS from the telecommunications provider. The new service extension experienced considerable growth during the analysis time frame. Most subscribers to the NTMS were adopting this technology for the first time. So although it was not a brand new service, it was "new" to nearly all subscribers during that time.

The data used to test the hypotheses were provided on an anonymous basis by a large U.S. telecommunications provider and included the following:

- Data collected as part of an ongoing nonbranded customer survey used to monitor consumer public perception of the company under study and its competitors. Responses were collected via random digit dialing over a 6-month period from survey households residing in five Midwestern states.
- Network data establishing the NTMS eligible customers that called, or were called by, the survey household (considered a "link" of the surveyed household) for the 3-month pre-survey period. Eligibility for NTMS is determined by classification as a residence in an area where the necessary infrastructure for accessing the NTMS was already in place.
- Account data establishing NTMS adoption by the eligible links, as well as the links' core service tenure.
- Profile variables for the survey households and eligible links, including service portfolio, tenure, and usage summary.
- Geo-level variables, including summarized census data and NTMS penetration as of the survey date.

Thus, the relevant data set for our study is a sample of survey households with customers of a telecommunications provider who use the firm's core service and the NTMS and links of these households that use the firm's core service but not the NTMS. Note that preliminary analyses revealed that links that are not already customers of the core service are extremely unlikely to adopt NTMS due in many cases to barriers of entry rather than any kind of attitudinal predisposition against the NTMS or the brand in general.

The final sample consisted of 791 survey households with 11,552 links who are eligible for the new service. A demographic profile of the survey households is provided in Table 1.

Household-level information on links, other than network and account information, is not available.

The data include 383 adoption events for an overall link adoption rate of 3.32%. Over one third (34.1%) of survey households had at least one eligible link adopt NTMS during the 3-month follow-up period and more than 1 out of 10 (10.8%) had multiple links adopt NTMS.

The dependent variable in the analysis (GetsNTMS) is an indicator variable for whether a link adopted NTMS during the 5 months following completion of the household survey (GetsNTMS = 1 if NTMS is adopted and 0 otherwise). Respondents to the survey question: "Would you actively encourage a friend or family member to consider [this new product from this company] WITHOUT being asked to do so?" were provided five choices: Definitely Would, Probably Would, Might or Might Not, Probably Would Not, and Definitely Would Not. We used this full five-level categorical variable in our analyses and also used the indicator (0–1) variable Rec5 to represent the "top box" (Rec5 = 1 if the household responded "Definitely Would" and 0 otherwise). Other important survey household covariates included (a) the total minutes on the phone between

the survey household and all local links during the 3 month prior to the survey (HHtime) and (b) an indicator variable for those households that had been using NTMS for no more than 24 months at the time of the survey (Tenure24). Tenure24 is used to identify “new adopters” among the potential influencers. Approximately two thirds (67%) of survey households fall into this category. The 24-month break coincides with the standard timing of contract renewal. Thus, all survey households with tenure shorter than 24 months are first-time contract holders of the NTMS. NTMS tenure is appended from the service provider’s transactional database not self-reported. A continuous indicator of tenure in months was tested in preliminary models but was not statistically significant. This finding is consistent with our expectation that the effect of tenure is non-linear. In addition to the 24-month break that coincides with first contract renewal, we tested alternate definitions of “new” customers at 12 months and 18 months. Below 12 months, we do not see sufficient adoption events to support significant findings. Versions of the final model however using these alternate breaks (12 and 18 months) were consistent with the findings using the 24-month definition.

To protect the privacy of individuals, only anonymous data were used in this study; no information that could be used to directly or indirectly identify individuals was made available to researchers. Customer account numbers were replaced with anonymous IDs via a hashing function. Survey responses, profile variables, and demographic detail were aggregated into relevant groups before access by researchers.

The most important covariates for the household links were (a) the number of calls made by the link during the 1 month prior to the survey (LinkCalls), (b) the percentage of customers in the links’ service region for whom NTMS is available, which is used as a proxy for NTMS availability at the link level (Availability), (c) the link’s core service tenure as a customer of the company, appended from the company’s transactional database and measured in months (LinkTenure), and (d) an indicator variable, StrongLink, which is “1” only if the link accounted for at least 5% of total caller time (all time spent on calls from links to household) and 5% of total callee time (total time spent on calls from household to links) time. When StrongLink = 1, there is a significant percentage of two-way communication between the link and household. This indicator variable performed better in the exploratory models than continuous univariate measures such as total time or percentage of total time spent on the phone between the link and survey household. Nine percentage of all links fall into this category (where StrongLink = 1). Table 2 provides descriptive statistics of all final model covariates.

Additional detail on the bivariate relationship between the model variables is provided in Appendices A and B.

Model Development

To accommodate the nested structure of the data (i.e., links within survey households) and the dichotomous nature of the outcome (i.e., link adopted NTMS vs. did not adopt), we used

a hierarchical generalized linear model for Bernoulli sampling. Here, Y_{ij} is “1” if link i of household j adopts NTMS and 0 otherwise and π_{ij} represents the corresponding probability of adoption, $\pi_{ij} = P[Y_{ij} = 1]$. Consequently,

$$E[Y_{ij}|\pi_{ij}] = \pi_{ij}, \text{Var}[Y_{ij}|\pi_{ij}] = \pi_{ij}(1 - \pi_{ij}).$$

Using the logit link function, $\eta_{ij} = \log(\pi_{ij}/(1 - \pi_{ij}))$, the link-level (or level 1) structural model for the log-odds, η_{ij} , is a linear function of the link characteristics $X_{ij}^{(1)}, \dots, X_{ij}^{(K)}$ (these include interactions between link characteristics and household characteristics),

$$\eta_{ij} = \beta_{0j} + \sum_{k=1}^K \beta_{kj} X_{ij}^{(k)}, \quad (1)$$

At Level 2, the intercept, β_{0j} , is a linear function of household j ’s characteristics $W_{1j}, \dots, W_{Lj}$ and random effect ξ_{0j} ,

$$\beta_{0j} = \gamma_{00} + \sum_{\ell=1}^L \gamma_{0\ell} W_{\ell j} + \xi_{0j}, \quad (2)$$

The household random effects, ξ_{0j} , are assumed to be multivariate normal with a common mean of zero and a diagonal covariance matrix Ξ (Raudenbush and Bryk 2002; Schabenberger and Gregoire 1996).

The random effects at the household level provide a flexible way to accommodate the natural dependence among links to the same household, and the two-level structure is ideal for examining cross-level effects, including the influence of a survey household on its multiple links. Consequently, we will differentiate between household-level effects (based on survey household characteristics) and link-level effects when assessing the relative impact of cross-level interactions on link-level outcomes. An alternative approach would be to use an aggregate model at the household level. Such a model would also accommodate the violation of the independence assumption at the link level, but it would not allow us to examine the effect of link-level characteristics on NTMS adoption rates. Another disadvantage of data aggregation is the dramatic decline in sample size and statistical power (Wieseke et al. 2008). Hierarchical linear models have been used recently to study many types of multilevel marketing phenomena (e.g., Inman, Winer, and Ferraro 2009; Rouziès et al. 2009; Wieseke et al. 2009).

Candidate variables for the model included telephone usage profile, service portfolio, and tenure of both survey household and links as customers; pre-survey communication between survey household and link; neighborhood availability and penetration of NTMS for both survey household and links; self-reported demographics for the survey household and census demographics for the links; and various attitudinal measures of the survey household, including likelihood to recommend and satisfaction with both NTMS and other services. The initial consideration set consisted of nearly 200 attributes. We used exploratory binary logistic stepwise regressions on the link-level data to identify the most promising potential predictors for the hierarchical model and a backward stepwise procedure to arrive at the final set of model covariates. “Model 0” in Table 3 represents the best model (using

Table 2. Descriptive Statistics of Model Variables for 791 Households and 11,552 Links

Variable	Description	Mean	Standard Deviation	1st Quartile	Median	3rd Quartile
Link-level Outcome						
GetsNTMS	Indicator variable for whether link acquires NTMS	3.3%	18%	0	0	0
Survey household-level covariates						
HHtime	Total Time (minutes) communicating with all local links based on 3 months pre-survey	2598	3252	1287	2359	4407
Rec5	Indicator that household "Definitely Would" recommend NTMS	37%	48%	0	0	1
Tenure24	Indicator for household that has had NTMS no longer than 24 months	68%	47%	0	1	1
Link-level covariates						
LinkCalls	Number of all long distance calls made by the link during 1 month pre-survey	21	43	3	9	23
Availability	Percentage of residential lines where NTMS is available	78%	16%	71%	83%	89%
LinkTenure	The number of months that the link has had current service with the firm	172	150	33	141	306
StrongLink	Indicator for when Caller and Callee times are both > 5% of corresponding household totals based on 3 months pre-survey	9%	28%	0	0	0

Note: NTMS = new-to-market service.

Bayesian Information Criterion [BIC]) that does not include the test variables. This model is used as the baseline model in the tests of the hypotheses.

Model Diagnostics

Diagnostic checks indicate that there are no significant problems with the underlying assumptions of the baseline model. In particular, the generalized goodness-of-fit chi-square test for homogeneity (Schabenberger 2005, p. 4) shows there is no significant residual overdispersion ([Generalized Chi-Square]/DF = 1.10 in the baseline model). To check the generalized linear structure of Expressions 1 and 2, we calculated the average observed and predicted adoption rates for each household's links and calculated residuals as the difference in the empirical logit of these observed and predicted averages by household. There was no significant correlation between these residuals and the empirical logit of the average prediction by household ($p > .1$), which supports the hierarchical linear structure of Expressions 1 and 2.

Finally, we studied several latent class analyses to determine if there was significant heterogeneity across various segments of the sample. Initially we considered segments that could be identified by adoption rates, without the explicit use of covariates, but the single-segment baseline model in Table 3 was better (using BIC) than all latent class alternatives. In subsequent analyses, we used household and link characteristics as candidate covariates for possible alternative segmentation models, but none of these covariates were statistically significant in alternative multiclass models.

Results

The results are summarized in Table 3. In the baseline model (Model 0), greater call volume (in terms of total survey

household minutes or total link number of calls) is associated with a greater likelihood of NTMS adoption. Also, links that have been customers for a shorter period (LinkTenure) are significantly more likely to adopt NTMS than those with longer tenure (*ceteris paribus*).

Perhaps conspicuous by its absence in the baseline model is any measure of homophily, or degree of similarity, between the links and survey households (Brown and Reingen 1987; Nitzin and Libai 2010). While several demographic attributes were available on the survey households, demographic data on links were limited to geo-level averages. Nevertheless, we were able to calculate a demographic similarity measure comparing survey households and links on income (median income for links), education (percentage college graduates), age (median age of head of household), household size (median household size), and race (percentage White).¹ When added to the baseline model, this attribute was not statistically significant. We also tested a dichotomous indicator of whether the link and survey household live in the same geographic area, as well as differences in tenure (absolute difference in months) for the existing service. Neither of these measures was significant individually or in any combination with the demographic measure ($p > .2$ in every case).

In Table 3, Models 1–4 address Hypotheses 1–4, respectively. Model 5 also addresses Hypothesis 4. Model 1 adds the variable, Rec5, which is the indicator for those households that "definitely would" recommend NTMS to friends and family. At this broad level, recommend intention has no significant impact on a link's likelihood to adopt NTMS by itself. Alternative versions of "Likelihood to Recommend" were tested as well, including the original 5-point scale as an ordinal predictor, a combination of dummy indicators for top box and bottom two, as well as the empirical logit transformation of the scale. We also tested various interactions between recommend intention and overall satisfaction with the service provider. None of

Table 3. Hierarchical Logistic Regression Models for Whether a Link Acquires New-to-Market Service (NTMS) Based on Household Recommend Intention and Covariates for Usage Tenure, and Level of Communication

Variable	Model 0	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept (standard error)	-4.601* (.332)	-4.571* (.336)	-4.578* (.351)	-4.545* (.337)	-4.593* (.333)	-4.551* (.351)
Survey household						
HHtime						
Rec5	3.1E-5* (.000012)	3.1E-5* (.000012)	3.1E-5* (.000012)	3.2E-5* (.000012)	3.2E-5* (.000012)	3.2E-5* (.000012)
Tenure24		-0.063 (.108)	-0.096 (.201)	-0.118 (.114)	-0.044 (.209)	-0.100 (.202)
Rec5 × Tenure24			0.010 (.145)		.069 (.152)	0.008 (.145)
Link level			0.045 (.239)		-.107 (.250)	-0.034 (.243)
LinkCalls	0.002* (.00076)	0.002* (.00076)	0.002* (.00076)	0.002* (.00076)	0.002* (.00076)	0.002* (.00076)
Availability	1.731* (.398)	1.724* (.398)	1.725* (.398)	1.722* (.398)	1.725* (.398)	1.725* (.398)
LinkTenure	-0.002* (.00038)	-0.002* (.00038)	-0.002 (.00038)	-0.002* (.00038)	-0.002* (.00038)	-0.002* (.00038)
StrongLink				-0.285 (.263)	.184 (.383)	-0.314 (.248)
Cross-level interactions						
Rec5 × StrongLink				0.606 (.373)	-.706 (.827)	
StrongLink × Tenure24					-.787 (.531)	
(Rec5 × Tenure24) × StrongLink					1.826† (.949)	0.832* (.379)
BIC	3353	3362	3381	3388	3412	3395
Generalized chi-square	11593.93	11575.78	11523.02	11542.36	11509.82	11505.15
Generalized chi-square/DF	1.0	1.0	1.0	1.0	1.0	1.0

Note: * $p < .05$. †One-tailed p value for directional hypothesis is significant at .05 level.

these alternatives was significant in the model. Thus, Hypothesis 1 is not supported.

In Model 2, we add two household-level covariates to Model 1: Tenure24 and an interaction term $\text{Rec5} \times \text{Tenure24}$ (this is an indicator for those households in the highest recommend category that have also acquired NTMS sometime in the last 24 months). Neither the interaction term, nor either of the main effects, is significant. Thus, Hypothesis 2 is not supported.

To test Hypothesis 3, we add three covariates to the baseline model: Rec5, StrongLink, and the interaction term for $\text{Rec5} \times \text{StrongLink}$. Again, these are all indicator variables. And again, we find that neither the interaction term nor the main effects has a significant impact on NTMS adoption. The cross-level interaction of $\text{Rec5} \times \text{StrongLink}$ is presented in Table 3 as a fixed effect, but we also tested it as a random effect. In each case, Hypothesis 3 is not supported.

Model 4 involves specification of a Level 2 interaction effect ($\text{Rec5} \times \text{Tenure24}$ from Model 2), which acts as a main effect for the cross-level interaction with StrongLink ($\text{Rec5} \times \text{Tenure24} \times \text{StrongLink}$). This is a fully specified model including all lower order interactions ($\text{Rec5} \times \text{StrongLink}$ and $\text{StrongLink} \times \text{Tenure24}$) and main effects. None of the main effects or low-order (i.e., two-way) interactions are statistically significant, but the coefficient for $\text{Rec5} \times \text{Tenure24} \times \text{StrongLink}$ is significantly positive ($p = .027$) and this supports the higher adoption rate that is described in Hypothesis 4. If the statistically insignificant coefficients of the lower order interactions and main effects are interpreted as being equal to zero, then based on the coefficient of $\text{Rec5} \times \text{Tenure24} \times \text{StrongLink}$ (1.826), links meeting this condition are 6.2 times as likely to adopt NTMS. The net effect of the three-way interaction when incorporating the coefficients of all constitutive elements is a 0.436 increase in the log-odds, which equates to a 1.5 times greater likelihood of adoption than links meeting none of these conditions. Thus, in addition to statistical significance, we find that the higher adoption rate described in Hypothesis 4 is substantial.

In Model 5, we seek to improve the fit of Model 4 by removing the two-way interaction effects $\text{StrongLink} \times \text{Tenure24}$ and $\text{Rec5} \times \text{StrongLink}$. Gill (2001) argues that higher order interaction terms between a lower order (e.g., two-way) interaction and a main effect may be interpreted the same way as any other two-way interaction. Since $\text{Rec5} \times \text{Tenure24}$ describes a condition of the survey household, while StrongLink describes a condition of the link, we view the $\text{Rec5} \times \text{Tenure24}$ distinction as a moderator of the StrongLink effect on the link's likelihood of adoption. When interpreted this way, the lower order interactions of $\text{StrongLink} \times \text{Tenure24}$ and $\text{StrongLink} \times \text{Rec5}$ become conceptually superfluous. Also, a comparison of the BIC values for each model indicates that Model 5 is substantially more plausible than Model 4 (the difference in BIC values exceeds 16, indicating that the odds favoring Model 5 over Model 4 are on the order of well over one thousand to 1, assuming equal prior odds; Kass and Raftery 1995).

While producing a more conceptually parsimonious and statistically better-fitting model, the absence of any lower order

interaction terms nonetheless has the potential to bias the coefficient of the higher order interaction. Nevertheless, Brambor, Clark, and Golder (2006) argue that it is permissible to remove lower order interactions from the model if they are established to be statistically indistinguishable from zero and the coefficient of the higher order interaction is sufficiently large in comparison to the coefficients of the lower order effects, as is the case in Model 4. Consequently, Model 5 provides an important alternative way of studying adoption rates, and in this model the coefficient for $\text{Rec5} \times \text{Tenure24} \times \text{StrongLink}$ is significantly positive ($p = .014$). Thus, we again find support for Hypothesis 4. Among households that are new adopters, who are also willing to recommend NTMS, this coefficient indicates that a link is approximately 2.3 times more likely to adopt NTMS when the relationship between the influencer and the potential new user is strong versus the case where the relationship is not strong (i.e., $\exp(.832) = 2.3$, where .832 is the coefficient of $\text{Rec5} \times \text{Tenure24} \times \text{StrongLink}$ in the linear Equation 2 for the log-odds of adoption; note that $\text{Rec5} \times \text{Tenure24}$ is also in this model). When incorporating coefficients for all constitutive elements (all of which are statistically indistinguishable from zero), we again see a 1.5 times greater likelihood of adoption for links that meet this criterion.

Although our three control measures for unobserved similarities between the links and survey households were not significant when added to the baseline model (Model 0, Table 3), there is still the question of whether the effect of Hypothesis 4 is driven primarily by correlated unobservables. Consequently, these three similarity measures were also added to the final models (Models 4 and 5, Table 3). The resulting models demonstrate that they have virtually no impact on the size and significance of $\text{Rec5} \times \text{Tenure24} \times \text{StrongLink}$ (which is significantly positive in every case, $p < .027$) and that none of the coefficients of these measures are significantly different from zero ($p > .2$ in each case).² Hypothesis 4 is also supported when the similarity measures are added to the model individually or in any combination.

We have defined "new adopters" as those who have a tenure that does not exceed 24 months, because that is the period before the first contract renewal. Although we do not see a sufficient number of adoption events to support significant findings at periods below 12 months, the final model (Model 5) does show the same significant effect supporting Hypothesis 4, if one defines a "new adopter" as one who has a tenure of no more than 12 or no more than 18 months (instead of 24). This is true whether or not one includes the similarity measures in the model. Taken together, these models indicate a likelihood of adoption that is between 1.5 and 2.7 times greater for links falling into the category defined by Hypothesis 4, and in every case the coefficient of the effect is significantly positive ($p < 0.014$, across each tenure period, and with and without the inclusion of similarity measures).³

The adoption rates for each level of categorization, in Table 4, provide additional, concrete illustrations of the effects demonstrated by the model coefficients. The last row of this table also

Table 4. New-to-Market-Service (NTMS) Link-Level Adoption Rates by Survey Household Recommendation, Tenure, and StrongLink

	N	Adoption Rate
REC5 (Link's survey household recommends NTMS)		
No	6,975	3.4%
Yes	4,577	3.2%
StrongLink (Caller and Callee times are both > 5% of corresponding household totals)		
No	10,534	3.3%
Yes	1,018	3.2%
Tenure24 (Link's survey household has NTMS tenure of 24 months or less)		
No	3,593	3.1%
Yes	7,959	3.4%
Interactions		
REC5 × Tenure24	3,221	3.4%
Stronglink × Tenure24	694	3.3%
REC5 × Stronglink	407	4.2%
REC5 × Stronglink × Tenure24	296	5.1%

shows that under the conditions of Hypothesis 4, the adoption rate is 5.1%, which is more than 1.5 times the base rate of 3.3% (shown in Table 2).

Discussion and Implications

Because of the potential of network-based marketing efforts created by social networks, WOM marketing efforts and WOM organizations have seen tremendous growth (Carl 2006; Vranica 2006; WOMMA 2007). This has led managers to place increasing—sometimes paramount—importance on recommend intention as a means of gauging adoption, customer loyalty and as a measure of expected future growth (Keiningham et al. 2007b; Reichheld 2003). The results of this study demonstrate, however, that recommend intention by itself has no significant impact on a potential customer's likelihood to adopt a NTMS brand extension in the industry and service investigated by this study. Furthermore, a customer has to be a more recent adopter of the product and in more frequent contact with potential customers in their network for WOM intentions to have an impact on adoption of the service extension. As noted earlier, often the impact of a predictor variable is only visible when examining its interaction with another variable (Preacher, Curran, and Bauer 2006). Under the conditions of Hypothesis 4, the highest rate of new subscriptions will increase by a factor of more than twice the adoption rate that prevails when the relationships between the household and link is not strong and over 1.5 times the base rate of 3.3%. Given the size of the market for this service, this represents a very large amount of sales revenue to the firm.

Given managers' current emphasis on tracking recommend intention as a means of predicting future growth (Keiningham et al. 2007b; Morgan and Rego 2007; Reichheld 2003), this finding has strong managerial implications for efforts to

measure and manage WOM intention, particularly for the adoption of a NTMS brand extension within a social network. The results indicate that measuring WOM intentions by itself may not be predictive in service adoption by other potential customers. It is clear that other factors need to be taken into consideration and measured when using this particular metric.

Our findings, however, point to the difficulty in translating social networking, recommend intention, and service adoption into a coherent marketing effort. First, and perhaps most significant, none of the expected predictor variables—recommend intention, recency of adoption, and strength of social connection (as measured by communication time)—demonstrated any statistically significant relationship with service adoption within the social network investigated when examined individually. Only by examining the interaction of these variables was it possible to identify a relationship to service adoption. Clearly, this finding illustrates the need for identifying meaningful segments that are most likely to influence adoption of NTMS brand extensions. Our research indicates that the focus needs to be on more recent adopters of the NTMS and those that interact more frequently with others in their network. This would appear to be supported by the findings of Godes and Mayzlin (2004) who found that the expected additional sales resulting from the WOM activities of loyal customers did not create anticipated additional sales, but WOM from non-loyal customers did result in increased sales (Godes and Mayzlin 2004; Yu 2005). It could well be that it is the “newness” of the WOM activity that has the greatest influence, as would be the case when new users, or non-loyal users, recommend the product/service.

Nevertheless, our results show that the effect only extended to the closest of connections. This also implies that the proximity of connections can be encouraged where possible through marketing efforts provided the investment is deemed worth the additional sales revenue from the service adoption by other customers. The results indicate that it is useless to have an audience without a persuasive WOM message but it is equally useless to have a great WOM message without an audience to listen to it. The nature of these two components is therefore found to be quite complementary.

Finally, social networks and WOM marketing efforts have become increasingly popular with marketers who introduce service extensions. As noted earlier, Taylor (2004) reports that 80% of marketing directors choose brand extensions as the main means of launching innovations. With regard to brand extensions and communication, research has largely focused on a company's communication of aspects of the brand extension (fit, benefits, etc.) to current customers of the parent brand (e.g., Bridges, Keller, and Sood 2000). While numerous researchers allude to WOM effects resulting from brand extensions (e.g., Diamantopoulos, Smith, and Grime 2005; Völckner, Sattler, and Kaufmann 2008), few have explicitly studied the subject directly. The notable exception is work done by Sjödin (2008), which investigated customers' inclinations to spread negative WOM about a brand extension. This research also contributes to the branding literature by examining some

of the factors that may impact success for one such NTMS brand extension.

Limitations and Future Research

As with all scientific research, there are limitations that need to be stated. Our research covered one NTMS brand extension in the telecommunications sector. While brand extensions are one of the most researched areas in marketing (Czellar 2003; Keller and Lehmann 2006; Völckner and Sattler 2006), other sectors and other services may differ in terms of the impact of recommend intention, recency of adoption, and connection strength on the diffusion of a service brand extension within a social network. Furthermore, the dynamics of adoption for entirely NTMS are possibly different than those for a service brand extension. While the advantage of examining service brand extensions is the popularity of extensions as a marketing strategy when introducing new services, it overlooks other means of new service introductions. This is a limitation that needs to be considered when interpreting results.

Also, it is important to recognize that although the impact of the interaction is statistically significant and managerially relevant, it still represents a relatively small absolute rate of adoption. It indicates that other factors may play a large role in the adoption of many, if not most, service brand extensions.

Also, our study examined social networking as it related to voice communication via the telephone. While this represents by far the largest electronic means of WOM communication and social networking, important differences exist between this form of social networking and Internet-based social networks (e.g., MySpace, Facebook, etc.). Therefore, it is possible that our findings would differ if they were conducted using Internet-based social networks.

Furthermore, there are certain limitations inherently associated with collaborative research. While the data in this research provide a very rich and real representation of actual customer behavior, it is also constrained to the data at hand. While we were able to examine the relationship between survey household and link in terms of quantity of communication (as

reflected in telecommunications records), we did not have access to link-level attitudinal information that would allow us to categorize the *attitudinal* similarity between survey and link households. After reviewing the list of candidate variables, however, we were able to identify five demographic attributes (age, household size, income, education, and race) for which we had geo-level data on links and household-level information on the survey households to measure similarity or homophily. We used a principal components analysis to create a factor based on these five attributes and then scored the survey households based on their household characteristics and the links based on their wirecenter (roughly equivalent to a town) averages and calculated the absolute value of the difference between the survey household and link factor scores and called this measure “homophily.” We were also able to test an indicator for whether the link and survey household reside in the same wirecenter, as well as controlling for differences in existing service tenure. None of these indicators were statistically significant when added to the baseline model, either individually or all together. When added to the final model, the inclusion of the similarity measures did not lead to any substantive changes. The three similarity measures tested against the baseline model were therefore able to offer us the best available controls for correlated unobservables.

Despite limitations, we believe that these results present compelling evidence of the importance of clearly identifying the necessary interaction effects for successfully marketing a NTMS brand extension via WOM within a social network. Furthermore, these results indicate a need to incorporate actual behavioral data in addition to customers’ attitudinal survey information when seeking to best understand the diffusion of service brand extensions within a social network. The trouble is that most good intentions often remain just that—good intentions. It is clear that at times, recommend intention is easier to collect compared to actual WOM behavior. The results of this research indicate however that when it is not possible to collect actual behavioral WOM data, it is important to include other customer metrics to gauge the real impact of intentions on actual new service adoption.

Appendix A

Table A1. Model Variables Covariance Matrix

Level 2 (Survey HH) Covariance Matrix								
	HHTime	Rec5	Tenure24					
HHTime	10574651.18	169.65	121.68					
Rec5	169.65	0.23	0.02					
Tenure24	121.68	0.02	0.22					
Level 1 (Links) Covariance Matrix								
	GetsNTMS	HHTime	Rec5	Tenure24	LinkCalls	Availability	LinkTenure	StrongLink
GetsNTMS	0.03	-22.18	0.00	0.00	-0.23	0.00	1.11	0.00
HHTime	-22.18	13719510.69	185.41	208.33	5160.87	46.07	-45186.59	-47.31
Rec5	0.00	185.41	0.24	0.01	-0.23	0.00	-1.31	0.00
Tenure24	0.00	208.33	0.01	0.21	0.01	0.00	-1.01	0.00
LinkCalls	-0.23	5160.87	-0.23	0.01	1863.58	-0.16	-658.42	0.00
Availability	0.00	46.07	0.00	0.00	-0.16	0.03	0.33	0.00
LinkTenure	1.11	-45186.59	-1.31	-1.01	-658.42	0.33	22452.31	-0.54
StrongLink	0.00	-47.31	0.00	0.00	0.00	0.00	-0.54	0.08
Level 2/Level 1 Covariance Matrix								
	HHTime	Rec5	Tenure24					
GetsNTMS	-22.18	0.00	0.00					
LinkCalls	5160.87	-0.23	0.01					
Availability	46.07	0.00	0.00					
LinkTenure	-45186.59	-1.31	-1.01					
StrongLink	-47.31	0.00	0.00					

Appendix B

Table B1. Model Variables Correlation Matrix

Level 2 (Survey HH) Correlation Matrix								
	HHTime	Rec5	Tenure24					
HHTime	1.00	0.11	0.08					
Rec5	0.11	1.00	0.07					
Tenure24	0.08	0.07	1.00					
Level 1 (Links) Correlation Matrix								
	GetsNTMS	HHTime	Rec5	Tenure24	LinkCalls	Availability	LinkTenure	StrongLink
GetsNTMS	1.00	-0.03	0.00	-0.01	-0.03	-0.04	0.04	0.00
HHTime	-0.03	1.00	0.10	0.12	0.03	0.08	-0.08	-0.05
Rec5	0.00	0.10	1.00	0.03	-0.01	-0.02	-0.02	0.00
Tenure24	-0.01	0.12	0.03	1.00	0.00	0.02	-0.01	0.00
LinkCalls	-0.03	0.03	-0.01	0.00	1.00	-0.02	-0.10	0.00
Availability	-0.04	0.08	-0.02	0.02	-0.02	1.00	0.01	0.00
LinkTenure	0.04	-0.08	-0.02	-0.01	-0.10	0.01	1.00	-0.01
StrongLink	0.00	-0.05	0.00	0.00	0.00	0.00	-0.01	1.00
Level 2/Level 1 Correlation Matrix								
	HHTime	Rec5	Tenure24					
GetsNTMS	-0.03	0.00	-0.01					
LinkCalls	0.03	-0.01	0.00					
Availability	0.08	-0.02	0.02					
LinkTenure	-0.08	-0.02	-0.01					
StrongLink	-0.05	0.00	0.00					

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Notes

1. The homophily measure was calculated by running a principal components analysis on the links' geo-level data. The survey households were then scored with the factor algorithm based on their household data. For example, the scoring coefficient for "percent white" was applied to the dummy variable for "white", where the household was either 100% or 0%. Similarly, the coefficient for median household income was applied to the survey household's self-reported income; the coefficient for median age of head of household was applied to respondent age, etc. We then took the absolute value of the difference of the two factors as our measure of similarity, where smaller differences indicated greater homophily.
2. $\text{Log-Odds of NTMS Adoption} = -4.723 + 0.000032 \text{ HHtime} (.365) - 0.09722 \text{ Rec5} (.000012) + 0.01322 \text{ Tenure24} (.145) - 0.04069 (\text{Rec5} \times \text{Tenure24}) (.243) + 0.001888 \text{ LinkCalls} (.00076) + 1.7717 \text{ Availability} (.403) - 0.00155 \text{ LinkTenure} (.000415) - 0.3144 \text{ StrongLink} (.248) + 0.8263 (\text{Rec5} \times \text{Tenure24} \times \text{StrongLink}) (.379) - 0.00672 \text{ Homophily} (.078) \times 0.1271 \text{ Samewire} (.107) + 0.000011 \text{ TenureDiff} (.000469)$
(Standard errors in parentheses).
3. Versions of the final model using these alternate breaks (12 and 18 month) were consistent with the findings using the 24 month definition. The coefficient of $\text{Rec5XTenure18XStrongLink}$ is 0.894 with a net effect of .376 when incorporating the coefficients for the constitutive effects. The coefficient of $\text{Rec5XTenure12XStrongLink}$ is 0.992 with a net effect of .543.

References

- Aaker, David A. and Kevin L. Keller (1990), "Consumer Evaluations of Brand Extensions," *Journal of Marketing*, 54 (January), 27-41.
- Anderson, E. W. and Vikas Mittal (2000), "Strengthening the Satisfaction Profit-Chain," *Journal of Service Research*, 3(2), 107-120.
- Arndt, Johan (1967), "Word-of-mouth Advertising and Informal Communication," in *Risk Taking and Information Handling in Consumer Behavior*, D. F. Cox, ed. Boston, MA: Division of Research, Harvard University.
- Bandiera, Oriana and Imran Rasul (2006), "Social Networks and Technology Adoption in Northern Mozambique," *Economic Journal*, 116 (514), 869-902.
- Barlyn, Suzanne (2007), "Talk Is Cheap: Word-of-Mouth Advertising Can be Targeted, Inexpensive and Effective—If Done Well," *Wall Street Journal* (Eastern edition, Nov 26), R6.
- Bass, Frank (1969), "A New Product Growth Model for Consumer Durables," *Management Science*, 15 (5), 215-227.
- Batra, Rajeev and Michael Ray (1986), "Situational Effects of Advertising Repetition: The Moderating Influence of Motivation, Ability, and Opportunity to Respond," *Journal of Consumer Research*, 12 (March), 432-445.
- Belch, George E. (1982), "The Effects of Television Commercial Repetition on Cognitive Response and Message Acceptance," *Journal of Consumer Research*, 9 (June), 56-65.
- Berlyne, D. E. (1970), "Novelty, Complexity, and Hedonic Value," *Perception and Psychophysics*, 8, 279-286.
- Bell, David R. and Sangyoung Song (2007), "Neighborhood Effects and Trial on the Internet: Evidence from Online Grocery Retailing," *Quantitative Marketing and Economics*, 5 (4), 361-400.
- Brambor, Thomas, William Roberts Clark, and Matt Golder (2006), "Understanding Interaction Models: Improving Empirical Analyses," *Political Analysis*, 14, 63-82.
- Bridges, Sheri, Kevin L. Keller, and Sanjay Sood (2000), "Communication Strategies for Brand Extensions: Enhancing Perceived Fit by Establishing Explanatory Links," *Journal of Advertising*, 29 (Winter), 1-11.
- Brown, Jacqueline J. and Peter H. Reingen (1987), "Social Ties and Word-of-Mouth Referral Behavior," *Journal of Consumer Research*, 14 (December), 350-362.
- Burke, Raymond R. and Thomas K. Srull (1988), "Competitive Interference and Consumer Memory for Advertising," *Journal of Consumer Research*, 15 (June), 55-66.
- Burt, Ronald S. (1987), "Social Contagion and Innovation: Cohesion versus Structural Equivalence," *American Journal of Sociology*, 92 (6), 1287-1335.
- Cacioppo, John and Richard E. Petty (1979), "Effects of Message Repetition and Position on Cognitive Response, Recall and Persuasion," *Journal of Personality and Social Psychology*, 37 (January), 97-109.
- Calder, Bobby J. and Brian Sternthal (1980), "Television Commercial Wearout: An Information Processing View," *Journal of Personality and Social Psychology*, 37 (January), 173-186.
- Carl, Walter J. (2006), "What's All the Buzz About? Everyday Communication and the Relational Basis of Word-of-Mouth and Buzz Marketing Practices," *Management Communication Quarterly*, 19 (May), 601-634.
- Chevalier, Judith A. and Dina Mayzlin (2006), "The Effect of Word of Mouth on Sales: Online Book Reviews," *Journal of Marketing Research*, 43 (3), 345-354.
- Conley, Timothy G. and Christopher R. Udry (2005), "Learning About a New Technology: Pineapple in Ghana," working paper, Yale University.
- Czellar, S. (2003), "Consumer Attitude Toward Brand Extensions: An Integrative Model and Research Propositions," *International Journal of Research in Marketing*, 20 (March), 97-115.
- Danaher, Peter J. and Roland T. Rust (1996), "Determining the Optimal Return on Investment for an Advertising Campaign," *European Journal of Operational Research*, 95 (3), 511-521.
- De Bruyn, Arnaud and Gary L. Lilien (2008), "A Multi-Stage Model of Word of Mouth through Viral Marketing," *International Journal of Research in Marketing*, 25 (3), 151-163.

- Diamantopoulos, Adamantios, Gareth Smith, and Ian Grime (2005), "The Impact of Brand Extensions on Brand Personality: Experimental Evidence," *European Journal of Marketing*, 39 (1/2), 129-149.
- Feick, Lawrence F. and Linda L. Price (1987), "The Market Maven: A Diffuser of Marketplace Information," *Journal of Marketing*, 15 (January), 83-97.
- Frenzen, Jonathan and Kent Nakamoto (1993), "Structure, Cooperation and the Flow of Market Information," *Journal of Consumer Research*, 20 (December), 360-375.
- Fournier, Susan and David Glen Mick (1999), "Rediscovering Satisfaction," *Journal of Marketing*, 63 (October), 5-23.
- Gill, Jeff (2001), "Interaction Hierarchies in Generalized Linear Models," American Political Science Association Annual Meeting, San Francisco, September 30, 2001.
- Godes, David and Dina Mayzlin (2004), "Using Online Conversations to Study Word-of-Mouth Communication," *Marketing Science*, 23 (Fall), 545-560.
- and ——— (2008), "Firm-Created Word-of-Mouth Communication: A Field-Based Quasi-Experiment," *Marketing Science*, 28 (4), 721-743.
- and ——— (2009), "Firm-Created Word of Mouth Communication: Evidence from a Field Test," *Marketing Science*, 28 (4), 721-739.
- Goldenberg, Jacob, Barak Libai, and Eitan Muller (2001), "Talk of the Network: A Complex Systems Look at the Underlying Process of Word-of-Mouth," *Marketing Letters*, 12 (3), 211-223.
- Gupta, Sunil and Valerie A. Zeithaml (2006), "Customer Metrics and Their Impact on Financial Performance," *Marketing Science*, 25 (November/December), 718-739.
- Herr, Paul M., Frank R. Kardes, and John Kim (1991), "Effects of Word-of-Mouth and Product-Attribute Information on Persuasion: An Accessibility-Diagnosticity Perspective," *Journal of Consumer Research*, 17 (March), 454-462.
- Heskett, J. L., T. O. Jones, G. W. Loveman, W. E. Sasser, and L. Schlesinger (1994), "Putting the Service-Profit Chain to Work," *Harvard Business Review*, 72 (March/April), 164-174.
- Hill, Shawndra, Foster Provost, and Chris Volinsky (2006), "Network-Based Marketing: Identifying Likely Adopters via Consumer Networks," *Statistical Science*, 21 (2), 256-276.
- Hogan, John E., Katherine N. Lemon, and Barak Libai (2003), "Quantifying the Ripple: Word-of-Mouth and Advertising Effectiveness," *Journal of Advertising Research*, 44 (September/October), 271-280.
- Inman, Jefferey J., Russell S. Winer, and Rosellina Ferraro (2009), "The Interplay Among Category Characteristics, Customer Characteristics, and Customer Activities on In-Store Decision Making," *Journal of Marketing*, 73 (September), 19-29.
- Iyengar, Raghuram, Christophe Van den Bulte, and Thomas W. Valente (2011), "Opinion Leadership and Social Contagion in New Product Diffusion," *Marketing Science*. <http://mktsci.journal.informs.org/cgi/content/abstract/mksc.1100.0566v1>
- Kahneman, Daniel, Peter P. Wakker, and Rakesh Sarin (1997), "Back to Bentham? Explorations of Experienced Utility," *Quarterly Journal of Economics*, 112 (2), 376-405.
- Kass, R. E. and A. E. Raftery (1995), "Bayes Factors," *Journal of the American Statistical Association*, 90 (430), 773-795.
- Keiningham, Timothy L., Bruce Coolil, Lerzan Aksoy, Tor Wallin Andreassen, and Jay Weiner (2007a), "The Value of Different Customer Satisfaction and Loyalty Metrics in Predicting Customer Retention, Recommendation and Share-of-Wallet," *Managing Service Quality*, 17 (4), 361-384.
- , ———, Tor Wallin Andreassen, and Lerzan Aksoy (2007b), "A Longitudinal Examination of Net Promoter and Firm Revenue Growth," *Journal of Marketing*, 71 (July), 39-51.
- Keller, Kevin L. and Donald R. Lehmann (2006), "Brands and Branding: Research Findings and Future Priorities," *Marketing Science*, 25 (November/December), 740-759.
- Kumar, V., J. Andrew Petersen, and Robert P. Leone (2007), "How Valuable is Word of Mouth?" *Harvard Business Review*, 85 (October), 139-146.
- Mangold, W. Glynn, Fred Miller, and Gary R. Brockway, (1999) "Word-of-Mouth Communication in the Service Marketplace," *Journal of Services Marketing*, 13 (1), 73-89.
- Morgan, Neil A. and Lopo Leotte Rego (2006), "The Value of Different Customer Satisfaction and Loyalty Metrics in Predicting Business Performance," *Marketing Science*, 25 (September-October), 426-439.
- Nitzin, Irit and Barak Libai (2010), "Social Effects on Customer Retention," Tel Aviv University working paper.
- PQ Media (2007), "Word-of-Mouth Marketing Forecast 2007-2011," PQ Media's Alternative Media Research Series. <http://www.pqmedia.com/word-of-mouth-marketing-forecast-2007.html>
- Ranaweera, Chatura and Jaideep Prabhu (2003), "On the Relative Importance of Customer Satisfaction and Trust as Determinants of Customer Retention and Positive Word of Mouth," *Journal of Targeting, Measurement and Analysis for Marketing*, 12 (1), 82-90.
- Raudenbush, Stephen W. and Anthony S. Bryk (2002), *Hierarchical Linear Models: Applications and Data analysis Methods*. Thousand Oaks, CA: SAGE.
- Reichheld, Frederick F. (2003), "The One Number You Need to Grow," *Harvard Business Review*, 81 (December), 46-54.
- (2006), *The Ultimate Question: Driving Good Profits and True Growth*. Boston, MA: Harvard Business School Press.
- Reingen, Peter H. and J. B. Kernan (1986), "Analysis of Referral Networks in Marketing: Methods and Illustration," *Journal of Marketing Research*, 23 (4), 370-378.
- Rogers, Everett (1962), *The Diffusion of Innovation*. New York, NY: The Free Press.
- Rouziès, Dominique, Anne T. Coughlan, Erin Anderson, and Dawn Iacobucci (2009), "Determinants of Pay Levels and Structures in Sales Organizations," *Journal of Marketing*, 73 (November), 92-104.
- Rust, Roland T., Anthony J. Zahorik, and Timothy L. Keiningham (1995), "Return on Quality (ROQ): Making Service Quality Financially Accountable," *Journal of Marketing*, 59 (April), 58-70.
- , Valerie A. Zeithaml, and Katherine N. Lemon (2000), *Driving Customer Equity*. New York, NY: The Free Press.
- and Tuck Siong Chung (2006), "Marketing Models of Service and Relationships," *Marketing Science*, 25 (November/December), 560-580.

- Schabenberger, O. (2005), "Introducing the GLIMMIX Procedure for Generalized Linear Mixed Models," Paper 196-30, SUGI Proceedings of the SAS Institute.
- and T. G. Gregoire (1996), "Population-Averaged and Subject Specific Approaches for Clustered Categorical Data," *Journal of Statistical Computation and Simulation*, 54 (1-3), 231-253.
- Sharp, Byron (2008), "Net Promoter Fails the Test," *Marketing Research*, Winter, 28-30.
- Sjödin, Henrik (2008), "Upsetting Brand Extensions: An Enquiry Into Current Customers' Inclination to Spread Negative Word of Mouth," *Brand Management*, 15 (March), 258-271.
- Silverman, George (2001), "The Power of Word of Mouth," *Direct Marketing*, 64 (September), 47-52.
- Stang, D. J. (1975), "The Effects of Mere Exposure on Learning and Affect," *Journal of Personality and Social Psychology*, 31 (1), 7-13.
- Stephen, Andrew T. and Donald R. Lehmann (2009), "Why Do People Transmit Word-of-Mouth? The Effects of Recipient and Relationship Characteristics on Transmission Behaviors," Columbia University working paper.
- and Jonah Berger (2010), "Creating Contagious: How Item Characteristics and Network Characteristics Combine to Drive Social Epidemics," working paper, Columbia University. <http://marketing.wharton.upenn.edu/documents/research/Creating%20Contagious.pdf> (Accessed 16 March 2011).
- Strang, David and Nancy Brandon Tuma (1993), "Spatial and Temporal Heterogeneity in Diffusion," *American Journal of Sociology*, 99 (3), 614-639.
- Taylor, David (2004), *Brand Stretch: Why 1 in 2 Extensions Fail, and How to Beat the Odds*. Hoboken, NJ: John Wiley and Sons, 10.
- Thompson, Richard F. and William Alden Spencer (1966), "Habituation: A Model Phenomenon for the Study of Neuronal Substrates of Behavior," *Psychological Review*, 73 (1), 16-43.
- Van den Bulte, Christophe and Gary L. Lilien (2001), "Medical Innovation Revisited: Social Contagion versus Marketing Effort," *American Journal of Sociology*, 106 (5), 1409-1435.
- and Stefan Stremersch (2004), "Social Contagion and Income Heterogeneity in New Product Diffusion: A Meta-Analytic Test," *Marketing Science*, 23 (Fall), 530-544.
- and Stefan Wuyts (2007), *Social Networks and Marketing*. Cambridge, MA: Marketing Science Institute.
- Verhoef, Peter, Philip Hans Franses, and Janny C. Hoekstra (2001), "The Impact of Satisfaction and Payment Equity on Cross-Buying: A Dynamic Model for a Multi-Service Provider," *Journal of Retailing*, 77 (3), 359-378.
- , Gerrit Antonides, and Arnoud N. DeHoog (2004), "Service Encounters as a Sequence of Events," *Journal of Service Research*, 7 (1), 53-64.
- Villanueva, Julian, Shinjin Yoo, and Dominique M. Hanssens (2008), "The Impact of Marketing-Induced Versus Word-of-Mouth Customer Acquisition on Customer Equity Growth," *Journal of Marketing Research*, 45 (1), 48.
- Völckner, Franziska and Henrik Sattler (2006), "Drivers of Brand Extension Success," *Journal of Marketing*, 70 (April), 18-34.
- , Henrik Sattler, and Gwen Kaufmann (2008), "Image Feedback Effects of Brand Extensions: Evidence from a Longitudinal Field Study," *Marketing Letters*, 19 (June), 109-124.
- Vranica, Suzanne (2006), "Buzz Marketers Score Venture Dollars," *Wall Street Journal*, Eastern edition (January 13), A11.
- Wangenheim, Florian V. and Tomás Bayón (2004), "The Effect of Word-of-Mouth on Services Switching: Measurement and Moderating Variables," *European Journal of Marketing*, 38 (9/10), 1173-1185.
- and ——— (2007), "The Chain from Customer Satisfaction via Word-of-Mouth Referrals to New Customer Acquisition," *Academy of Marketing Science*, 35 (2), 233-249.
- Wieseke, Jan, Michael Ahearne, Son K. Lam, and Rolf van Dick (2009), "The Role of Leaders in Internal Marketing," *Journal of Marketing*, 73 (March), 123-145.
- , Nick Lee, Amanda J. Broderick, Jeremy F. Dawson, and Rolf Van Dick (2008), "Multilevel Analyses in Marketing Research: Differentiating Analytical Outcomes," *Journal of Marketing Theory and Practice*, 16 (4), 321.
- Wu, Cochen and Yung-Chien Yen (2007), "How the Strength of Parent Brand Associations Influence the Interaction Effects of Brand Breadth and Product Similarity with Brand Extension Evaluations," *The Journal of Product and Brand Management*, 16 (5), 334-341.
- Yu, Larry (2005), "How Companies Turn Buzz Into Sales," *MIT Sloan Management Review*, 46 (Winter), 5-6.
- WOMMA (2007), "Press Release: Word of Mouth Marketing is Here to Stay: WOMMA Reaches Membership Milestone, Looks Forward to Industry Growth," February 15. <http://www.womma.org/news/008905.php> (Accessed March 23, 2008).
- Zeithaml, Valerie A., Ruth N. Bolton, John Deighton, Timothy L. Keiningham, Katherine N. Lemon, and J. Andrew Peterson (2006), "Forward-Looking Focus: Can Firms Have Adaptive Foresight?" *Journal of Service Research*, 9 (November), 168-183.

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